

Eric Gerard Rothstein Morris

Contact

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Links: [LinkedIn](#), [GitHub](#), [Webpage](#)

Summary

I am a computer scientist with a Ph.D. in Information Systems. My research applied formal methods to diverse security domains: I used bounded model checking to generate and classify attackers, built LLVM compiler passes that reduce timing side-channel leakage, and defined metrics for integrity in cyber-physical systems security. Since 2022, I have worked in industry, most recently running Elastic SIEM clusters for German healthcare institutions with Ansible, Docker Swarm, and CI/CD. I am learning Lean 4 and want to build tools that make formally verified code the default when AI writes software.

Research Projects

ORIGAMI: Folding Data Structures to Reduce Timing Side-Channel Leakage (MEMOCODE 2022): I developed compiler passes in C++ for **clang**/LLVM, including a custom taint analysis. The passes standardise and transform LLVM's intermediate representation to follow constant-time programming practices.

Systematic Classification of Attackers via Bounded Model Checking (VMCAI 2020): I generated attackers for bit-level hardware models (And-inverter graphs) and classified them by the security requirements they can break, using SAT-based bounded model checking with **CaDiCaL**. I developed two sound heuristics, based on cone-of-influence reduction and monotonicity, and evaluated them on Hardware Model Checking Competition benchmarks.

Latent Behaviour Analysis of the Testbed of a Water Treatment Plant: I used **Python** to generate and study attacker strategies on a linear, time-invariant model of the Secure Water Treatment testbed (SWaT) at SUTD.

Employment

SIEM Engineer – März Internetwork Services AG

June 2024 – Present

Maintain, troubleshoot and deploy self-managed **Elastic Stack** clusters that run security information and event management (SIEM) systems. Security analysts use these systems to monitor and protect several healthcare institutions in Germany.

Currently refactoring the production SIEM infrastructure to introduce the **CI/CD** practices it lacked.

Automate infrastructure using **Ansible** and manage container orchestration with **Docker Swarm**.

Develop small containerised **Python** applications that use the Elasticsearch APIs to help security analysts handle alerts.

Data Engineer – Mapegy GmbH**January 2023 – March 2024**

Created data deployment processes in **PostgreSQL** and **Python** for innovation scouting with a focus on startups.

Software Quality Engineer (Freelance)**March 2022 – December 2022**

Built a **Python** tool that measures code coverage of **PostgreSQL** functions, which helped find dead code and untested paths.

Helped design new unit tests and improve existing ones for PostgreSQL functions.

Research Assistant – Universität Passau**September 2011 – September 2016**

Chair of IT Security (Prof. Joachim Posegga)

Explored a synthesis technique for the enforcement of security properties by solving behavioural equations in a coalgebraic setting (ESSoS 2016).

Worked on the EU FP7 project BIOMICS, where I prototyped a computational model inspired by cellular processes with a team of biologists, mathematicians, and physicists.

Academic Publications

E. Rothstein-Morris, J. Sun, and S. Chattopadhyay. **ORIGAMI: Folding Data Structures to Reduce Timing Side-Channel Leakage.** *20th ACM-IEEE International Conference on Formal Methods and Models for System Design, MEMOCODE 2022.*

E. Rothstein-Morris, J. Sun, and S. Chattopadhyay. **Systematic Classification of Attackers via Bounded Model Checking.** *21st International Conference on Verification, Model Checking, and Abstract Interpretation, VMCAI 2020.*

E. Rothstein-Morris, C. G. Murguía, and M. Ochoa. **Design-time Quantification of Integrity in Cyber-physical Systems.** *12th ACM SIGSAC Workshop on Programming Languages and Analysis for Security, PLAS 2017.*

E. Rothstein Morris and J. Posegga. **Idea: Enforcing Security Properties by Solving Behavioural Equations.** *8th International Symposium on Engineering Secure Software and Systems, ESSoS 2016.*

C. L. Nehaniv, J. Rhodes, A. Egri-Nagy, P. Dini, E. Rothstein-Morris, G. Horváth, F. Karimi, D. Schreckling, and M. J. Schilstra. **Symmetry structure in discrete models of biochemical systems: natural subsystems and the weak control hierarchy in a new model of computation driven by interactions.** *Philosophical Transactions of the Royal Society A, 2015.*

Education

Ph.D. in Information Systems Technology and Design**September 2016 – September 2021**

Singapore University of Technology and Design (SUTD)

Thesis: *Latent Behaviour Analysis*. Advisor: Sudipta Chattopadhyay.

Funded by the SUTD President's Graduate Fellowship.

M.Sc. in Engineering (Systems and Computing)**January 2009 – March 2011**

Universidad de los Andes, Colombia

Taught introductory courses in Programming Language Theory, Formal Methods, and Theory of Algorithms.

B.Sc. in Systems and Computing Engineering

January 2004 – December 2008

Universidad de los Andes, Colombia

Minor in Computational Mathematics.

Technical Skills

Programming languages: Python, C++, SQL, Lean 4 (currently learning).

Tools: Elastic Stack, Docker, Docker Swarm, Ansible, Git, PostgreSQL, LLVM/clang.

Languages: Spanish (Native), English (Professional - C1), German (Intermediate - B2).

References

Peter Walde
CEO at Mapegy GmbH

Sudipta Chattopadhyay
Associate Professor, School of Science and Engineering,
University of Missouri–Kansas City
(UMKC)

Jorge Cuellar
Former Principal Research Scientist at Siemens AG
Adjunct Professor at Universidad de Los Andes

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